Selective and Calibrated Decoding for Quantum Error Correction: A Topology-Aware Decoder-Family Benchmark under Structured Noise

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Abstract

A quantum error-correcting code protects information only if its *decoder* maps a measured syndrome to the right recovery under the *structured* noise that real hardware exhibits: bias, spatial bursts, neighbour correlations and faulty syndrome read-out. Most decoder studies optimise a single number, the logical error rate, under independent-and-identically-distributed (i.i.d.) noise. We argue that two further axes matter at least as much for deployable decoding yet are usually treated as afterthoughts: the ability to *abstain* when a decoder is unsure (selective decoding), and the *calibration* of the confidence that gates abstention. We build a unified, reproducible benchmark that evaluates a family of eight decoders—spanning optimal lookup, classical matching and belief propagation, and learned neural and reinforcement-learning selectors—across five structured-noise regimes on a distance-d repetition code, treating logical error rate, expected calibration error (ECE) and the risk–coverage trade-off as first-class metrics with analytic 95% confidence intervals. At our reported scale (distance 9; 40,000 trials per regime, 200,000 pooled shots per decoder), the decoders are *statistically indistinguishable* on raw logical error (means 0.1516–0.1540, 95% intervals $\approx \pm 0.0016$; the 0.0024 best-to-worst spread is below one interval width), so claims that one decoder "beats" another are not supported. The separable contributions lie elsewhere: (i) a framework comparing the whole family under structured noise on one defect-graph interface; (ii) abstention, which drives matching's committed logical error from 0.1529 at full coverage to 0.0757 at 0.55 coverage and 0.0099 at 0.25 coverage; and (iii) calibration, which varies eight-fold across decoders and is uncorrelated with sophistication—lowest for the lightweight neural selector (0.0317), worst for plain matching (0.2608), contradicting a natural intuition that the fused decoder is best calibrated. We report the artifact's numbers, not the intuition, and frame the work as a calibrated, selective-decoding benchmark, not a superiority claim.

1 Introduction

Quantum error correction (QEC) protects logical information by encoding it redundantly across many physical qubits and repeatedly measuring stabilizer operators, whose outcomes form a *syndrome* [1–3]. Topological codes—most prominently Kitaev's toric and surface codes [4–6]—are the leading candidates for scalable fault tolerance, and recent hardware has demonstrated that scaling such a code can suppress the logical error rate below the physical one [7]. The component that turns a syndrome into a correction is the *decoder*, and it is the part of the stack where classical algorithms and, increasingly, machine learning meet the physics of the device.

The dominant decoder for surface codes is minimum-weight perfect matching (MWPM) [5, 6, 8], built on Edmonds' blossom algorithm [9]; belief propagation (BP) and BP with ordered-statistics decoding underpin quantum low-density-parity-check decoding [10–12]. A large and

growing literature applies neural networks and reinforcement learning to decoding [13–18], and very recent work demonstrates learned decoders that exceed matching-based accuracy on real hardware syndromes [19]. The headline metric in nearly all of this work is the logical error rate, almost always evaluated under i.i.d. (depolarizing or bit-flip) noise.

Two facts motivate this paper. First, hardware noise is not i.i.d. It is *structured*: dominated by one Pauli type (bias), spatially bursty, correlated between neighbours, and accompanied by faulty syndrome extraction. A decoder that is excellent on i.i.d. noise may or may not retain its advantage under structure, and the only way to know is to test the same decoder family across regimes on one footing. Second, a decoder that runs inside a fault-tolerant control loop does not have to commit to every syndrome. It can *abstain*—flagging a shot as unreliable so that a slower fallback decoder, a repeated measurement, or a higher-level protocol takes over. Abstention is the QEC analogue of *selective prediction* in machine learning [20–22], and it is only useful if the confidence that gates it is *calibrated* [23, 24]. These two axes—coverage and calibration—are routinely omitted from decoder benchmarks.

This work. We treat decoding as a problem on the code's *defect graph* and build a unified, fully reproducible benchmark with three properties usually kept separate. (1) *Breadth*: a single `decode(syndrome)` → `(correction, confidence)` interface hosts eight decoders spanning the textbook-to-learned spectrum, evaluated across five structured-noise regimes. (2) *Selectivity*: every decoder can abstain, so we report risk–coverage curves, not just a single error rate. (3) *Calibration*: expected calibration error is a first-class output alongside the logical error rate, with analytic 95% confidence intervals so that statistical significance is visible rather than assumed.

What we claim, and what we do not. We are deliberately conservative. On the repetition code and the regimes we study, at reported scale the eight decoders' logical error rates lie within each other's confidence intervals (Sec. 3.2); we therefore do *not* claim that any decoder beats another on raw accuracy, and we say so explicitly. What the benchmark does establish is (i) that abstention buys a large, monotone reduction in committed logical error—matching's committed risk falls from 0.1529 at full coverage to 0.0757 at 0.55 coverage and 0.0099 at 0.25 coverage (Sec. 3.3)—and (ii) that calibration quality is *not* predicted by decoder sophistication: ECE ranges from 0.0317 (the lightweight neural coset selector) to 0.2608 (plain minimum-weight matching), an eight-fold spread with no monotone relationship to decoder complexity. The contribution is the framework, the selective/calibrated lens, and the honest accounting—not a new state-of-the-art error rate. This audit ethos is enforced in code: a readiness gate rejects any reported number that does not trace to a generated artifact and rejects forbidden absolute-superiority phrasing.

1.1 Contributions

1. **A decoder-family benchmark under structured noise.** Eight decoders (lookup, majority, matching, BP, neural, topology, fused, RL selector) on one defect-graph interface, evaluated across i.i.d., biased, burst, correlated, and measurement-error regimes, with per-regime breakdowns and threshold curves.

2. **Selective decoding as a primary axis.** Confidence-gated abstention and risk–coverage curves for every decoder, quantifying the reliability bought by giving up coverage.

3. **Calibration as a first-class metric.** Expected calibration error reported for all decoders with 95% confidence intervals, exposing that calibration is uncorrelated with decoder complexity.

4. **An honest, reproducible accounting.** A single source-of-truth artifact, analytic confidence intervals that reveal the decoders to be statistically indistinguishable on raw logical error, and a code-level audit gate against fabricated numbers and superiority claims.

2 Theory

This section makes precise, and proves in full, every structural claim that the benchmark relies on: the algebraic characterisation of the repetition syndrome and of logical failure (Sec. 2.1); the optimality of the lookup/coset-leader decoder and the syndrome-consistency of the matching, majority and learned decoders (Sec. 2.2); the exact equivalence between matching, majority and lookup on the line, which is the formal origin of the observed statistical tie (Sec. 2.3); automorphism invariance of the logical error rate (Sec. 2.4); the Bernoulli confidence interval and the indistinguishability criterion (Sec. 2.5); and the calibration–selectivity guarantees that justify reporting risk–coverage and ECE as first-class metrics (Sec. 2.6). Each result is tied to the experiment that illustrates it. The results are stated conservatively: they concern small, fully reproducible simulated benchmarks (a repetition code and a surface-like toy lattice) under simulated structured noise; they are not hardware syndromes, not a full surface code at scale, and not a claim of superiority over production decoders (Sec. 4.1).

2.1 The repetition code: syndromes and logical failure

We work over the binary field $\mathbb{F}_2 = \{0, 1\}$ with addition $\oplus$ (XOR) and the Hamming weight $|x| = \sum_i x_i$ (number of nonzero coordinates). Fix a distance $d \geq 2$.

Definition 1 (Repetition code)**.** *The distance-d repetition code has data space $\mathbb{F}_2^d$ and parity-check matrix $H \in \mathbb{F}_2^{(d-1)\times d}$ with rows H_i, $i = 0, \ldots, d-2$, given by $(H_i)_i = (H_i)_{i+1} = 1$ and all other entries 0. The* syndrome *of an error $e \in \mathbb{F}_2^d$ is $s(e) = He \in \mathbb{F}_2^{d-1}$, i.e. $s_i(e) = e_i \oplus e_{i+1}$. The* logical functional *is $L\colon \mathbb{F}_2^d \to \mathbb{F}_2$, $L(e) = \left(\sum_i e_i\right) \bmod 2$. The* stabilizer group *is $\mathcal{S} = \ker H = \{x : Hx = 0\}$. A correction $\hat{c} \in \mathbb{F}_2^d$ applied to error e* succeeds *iff the residual $r = e \oplus \hat{c}$ satisfies $Hr = 0$ and $L(r) = 0$, and is a* logical failure *otherwise.*

The map H is the boundary operator of the one-dimensional chain complex whose 1-cells are the d data bits and whose 0-cells are the $d-1$ links; the next lemma records the elementary facts that make the line picture exact.

Lemma 1 (Kernel, image and the endpoint structure of the syndrome)**.** *For the matrix H of Definition 1:*

(i) $\ker H = \{0, \mathbf{1}\}$*, where $\mathbf{1} = (1, \ldots, 1)$; in particular $\dim_{\mathbb{F}_2} \ker H = 1$ and $\mathcal{S} = \{0, \mathbf{1}\}$.*

(ii) H has rank $d-1$, so $s \mapsto e$ is surjective onto $\mathbb{F}_2^{d-1}$: every syndrome is realisable.

(iii) For any error e, the set $J = \{i : s_i(e) = 1\}$ of lit checks is exactly the set of links i at which $e_i \neq e_{i+1}$, i.e. the boundaries *between maximal constant runs of e. Consequently $|J|$ is even iff $e_0 = e_{d-1}$, and the lit checks are the endpoints of the error chains.*

Proof. (i) If $Hx = 0$ then $x_i = x_{i+1}$ for all $i = 0, \ldots, d-2$, so x is constant; the only constant vectors in $\mathbb{F}_2^d$ are 0 and $\mathbf{1}$. Both lie in $\ker H$: $H0 = 0$ trivially, and $(H\mathbf{1})_i = 1 \oplus 1 = 0$. Hence $\ker H = \{0, \mathbf{1}\}$ and its $\mathbb{F}_2$-dimension is 1.

(ii) By rank–nullity over $\mathbb{F}_2$, $\operatorname{rank} H = d - \dim \ker H = d - 1$, which equals the number of rows, so the row space is all of $\mathbb{F}_2^{d-1}$ and H is surjective.

(iii) By definition $s_i(e) = e_i \oplus e_{i+1} = 1$ iff $e_i \neq e_{i+1}$, which is precisely a boundary between a maximal run of equal values ending at i and the next run starting at $i+1$. Writing e as an alternating sequence of maximal constant runs, each internal boundary toggles the value, so the

number of toggles between coordinate 0 and coordinate $d-1$ has the same parity as $e_0 \oplus e_{d-1}$; thus $|J| = \sum_i (e_i \oplus e_{i+1}) \equiv e_0 \oplus e_{d-1} \pmod 2$, which is even iff $e_0 = e_{d-1}$. Each "1-run" of e contributes a lit check at each of its two ends (or a single lit check at an open boundary of the line), i.e. the lit checks mark chain endpoints. □

Lemma 1 turns decoding into pairing endpoints on a line, the picture used by every defect-graph decoder in Sec. 5.3 and drawn in Fig. 1. The next proposition is the exact characterisation of "the code protected the bit" implemented in the runner.

Proposition 2 (Cosets and the failure dichotomy)**.** *Two errors e, e' have the same syndrome iff $e \oplus e' \in \{0, \mathbf{1}\}$. Hence each syndrome s labels exactly one* coset $e_s \oplus \mathcal{S}$ *of size 2, whose two representatives have logical values $L(e_s)$ and $L(e_s) \oplus (d \bmod 2)$. A correction $\hat{c}$ with $H\hat{c} = s(e)$ succeeds iff $L(\hat{c}) = L(e)$; a correction with $H\hat{c} \neq s(e)$ always fails.*

Proof. $He = He'$ iff $H(e \oplus e') = 0$ iff $e \oplus e' \in \ker H = \{0, \mathbf{1}\}$ by Lemma 1(i). Thus the fibre $H^{-1}(s)$ is the coset $e_s \oplus \{0, \mathbf{1}\} = \{e_s,\, e_s \oplus \mathbf{1}\}$, a set of size two. Its two members differ by $\mathbf{1}$, and $L(e_s \oplus \mathbf{1}) = L(e_s) \oplus L(\mathbf{1}) = L(e_s) \oplus (d \bmod 2)$ because L is $\mathbb{F}_2$-linear and $L(\mathbf{1}) = \sum_i 1 = d \bmod 2$.

For the failure dichotomy, let $r = e \oplus \hat{c}$. If $H\hat{c} \neq s(e) = He$ then $Hr = He \oplus H\hat{c} \neq 0$, so $r \notin \mathcal{S}$ and the correction fails by Definition 1. If $H\hat{c} = He$ then $Hr = 0$, i.e. $r \in \{0, \mathbf{1}\}$; the residual is then a stabilizer, and success requires additionally $L(r) = 0$. Since L is linear, $L(r) = L(e) \oplus L(\hat{c})$, so $L(r) = 0 \iff L(\hat{c}) = L(e)$. □

2.2 Decoder optimality and syndrome-consistency

Fix the i.i.d. bit-flip channel at physical rate $p < \frac{1}{2}$: $\Pr(e) = p^{|e|}(1-p)^{d-|e|}$.

Theorem 3 (The coset-leader (lookup) decoder is maximum-likelihood and minimises logical error)**.** *For $p < \frac{1}{2}$, the decoder that on syndrome s outputs a minimum-Hamming-weight representative of the fibre $H^{-1}(s)$ (a* coset leader*) is a maximum-likelihood decoder for the i.i.d. channel, and among all decoders it minimises the probability of logical failure on that channel.*

Proof. Because $p < \frac{1}{2}$ we have $0 < \frac{p}{1-p} < 1$, so $\Pr(e) = (1-p)^d \big(\frac{p}{1-p}\big)^{|e|}$ is a strictly decreasing function of $|e|$. Hence among all errors consistent with an observed syndrome s, i.e. all $e \in H^{-1}(s)$, the most likely is exactly the one of least weight; outputting a coset leader $\hat{c} = \arg\min_{e \in H^{-1}(s)} |e|$ therefore maximises $\Pr(e \mid s) \propto \Pr(e)\mathbf{1}[He = s]$. This is maximum likelihood.

For optimality of the *logical* error, condition on the observed syndrome s. By Proposition 2 the true error lies in a coset of size two with the two members carrying the two possible logical values; the residual is a stabilizer for *any* decoder that returns some $\hat{c} \in H^{-1}(s)$ (a decoder returning $\hat{c} \notin H^{-1}(s)$ fails with conditional probability 1 and is never better). Conditioned on s, success requires $L(\hat{c}) = L(e)$; the posterior probability of the logical class $\ell \in \mathbb{F}_2$ is $\Pr(L(e) = \ell \mid s) \propto \sum_{e \in H^{-1}(s),\, L(e) = \ell} \Pr(e)$. The Bayes-optimal rule picks the more probable logical class. The coset leader has weight no larger than the other representative (which differs by $\mathbf{1}$ and so has weight $d - |\text{leader}| \geq |\text{leader}|$ when the leader weight is $\leq d/2$, with equality only at $d/2$); since $\Pr(\cdot)$ is strictly decreasing in weight, the leader's logical class is the maximum-a-posteriori class, except in the measure-of-ties boundary case $|\text{leader}| = d/2$ where the two classes are exactly equiprobable and any choice is optimal. Thus the coset-leader rule realises the Bayes-optimal (minimum logical error) decision for every syndrome and hence minimises the overall logical failure probability. □

This is why `lookup` is the yardstick of Table 1: no decoder on the i.i.d. channel can beat it, which already caps how far any competitor's accuracy can be from it. We next show the heuristic decoders never violate the basic constraint, so the only freedom left is the logical bit and the confidence.

Proposition 4 (Syndrome-consistency of matching, majority and the learned decoders)**.** *On the line, each of the matching, majority, neural, topology and fused decoders of Sec. 5.3 returns a correction $\hat{c}$ with $H\hat{c} = s$ for every input syndrome s. Consequently, by Proposition 2, each fails on a given shot iff it selects the wrong logical class, never because of an unsatisfied syndrome.*

Proof. It suffices to exhibit, for each decoder, a correction in $H^{-1}(s)$. *Majority:* the cumulative XOR $c_j = \bigoplus_{i<j} s_i$ defines a chain with $c_j \oplus c_{j+1} = s_j$, i.e. $Hc = s$; the decoder returns this c or $c \oplus \mathbf{1}$, both in $H^{-1}(s)$ by Lemma 1(i). *Matching:* pairing the lit checks J into chains and flipping the data bits strictly between paired endpoints produces an error whose only toggles are at the elements of J (with a boundary closure supplying the missing endpoint when $|J|$ is odd, which by Lemma 1(iii) corresponds to $e_0 \neq e_{d-1}$); hence its syndrome is exactly s. *Neural/topology/fused:* by construction each starts from a syndrome-determined base chain $c_0 \in H^{-1}(s)$ (the same prefix chain as majority) and only ever adds a stabilizer $\in \{0, \mathbf{1}\}$ to choose the logical class, which does not change the syndrome since $H\mathbf{1} = 0$. Thus every output lies in $H^{-1}(s)$. The second claim is immediate from Proposition 2: with $H\hat{c} = s = He$, failure occurs iff $L(\hat{c}) \neq L(e)$. □

The neural decoder of Sec. 5.3 therefore reduces decoding to a single binary classification (the logical coset), which is why its confidence—a predicted class probability—is approximately calibrated (Sec. 3.4, Fig. 3).

2.3 Why the decoders tie: exact agreement on the line

Theorem 5 (Matching, majority and lookup coincide on the repetition line)**.** *On the distance-d repetition code, for every syndrome s the lookup, majority and matching decoders return corrections in the* same *coset $H^{-1}(s)$ and select the* same *logical class as the minimum-weight coset leader, except on the measure-zero set of syndromes whose coset leader has weight exactly $d/2$. Hence on the i.i.d. channel all three have identical logical error rate, and any decoder of Proposition 4 that breaks ties toward the lighter coset agrees with them.*

Proof. By Proposition 4 all three decoders output members of $H^{-1}(s)$, so they agree on the coset; only the logical class (equivalently, the choice between c and $c \oplus \mathbf{1}$) can differ. The lookup decoder picks the lighter of $\{c, c \oplus \mathbf{1}\}$ by definition. The majority decoder explicitly compares $|c|$ and $|c \oplus \mathbf{1}| = d - |c|$ and returns the lighter, so it picks the same representative whenever $|c| \neq d/2$. The matching decoder pairs the lit checks J left-to-right on the line; on a one-dimensional line the unique perfect pairing of ordered endpoints into non-crossing intervals minimises the total flipped length $\sum_{\text{pairs}}(j_{\text{right}} - j_{\text{left}})$, which equals the Hamming weight of the produced chain, so matching also returns a minimum-weight representative, i.e. the lighter of $\{c, c \oplus \mathbf{1}\}$, again whenever $|c| \neq d/2$. Therefore all three select the same logical class off the tie set $\{\, s : |\text{leader}(s)| = d/2 \,\}$, and on that set Theorem 3 shows the two classes are equiprobable so any choice has the same conditional failure probability. Equal per-syndrome conditional failure for every syndrome implies equal logical error rate on the i.i.d. channel. □

Theorem 5 is the analytic origin of the central empirical finding: the decoders are *statistically indistinguishable* on raw logical error (Table 1, Fig. 2); on the line they are, up to a negligible tie set and finite-sample noise, the *same* decision rule, so the 0.0024 best-to-worst spread is residual sampling fluctuation, not signal.

2.4 Automorphism invariance of the logical error rate

Definition 2 (Cyclic relabeling)**.** *For a shift $t \in \{0, \ldots, d-1\}$ let $\rho_t \colon \mathbb{F}_2^d \to \mathbb{F}_2^d$ be the cyclic shift $(\rho_t e)_i = e_{(i-t) \bmod d}$. A decoder $\mathcal{D}$ is* cyclically equivariant *if $\mathcal{D}(s(\rho_t e)) = \rho_t(\mathcal{D}(s(e)))$ for all e, t.*

Theorem 6 (Automorphism invariance)**.** *Let the noise distribution be cyclically invariant* ($\Pr(\rho_t e) = \Pr(e)$ *for all* t*), as holds for the i.i.d. channel. If a decoder* $\mathcal{D}$ *is cyclically equivariant, then its logical error rate is invariant under cyclic relabeling: for every shift* t*,* $\Pr_e[\mathcal{D} \text{ fails on } e] = \Pr_e[\mathcal{D} \text{ fails on } \rho_t e]$*.*

Proof. The logical functional is cyclic-invariant: $L(\rho_t e) = \sum_i e_{(i-t) \bmod d} = \sum_i e_i = L(e)$, since cyclic shift permutes the summands. The cyclic shift also commutes with the (cyclic) check structure, so $s(\rho_t e) = \pi_t\, s(e)$ for the induced shift π_t on links. Let $F(e) = \mathbf{1}[\mathcal{D} \text{ fails on } e]$. By equivariance $\mathcal{D}(s(\rho_t e)) = \rho_t \mathcal{D}(s(e))$, and the residual transforms as $\rho_t e \oplus \mathcal{D}(s(\rho_t e)) = \rho_t\big(e \oplus \mathcal{D}(s(e))\big)$. Since both H and L are cyclic-invariant in the sense above, the residual of the shifted problem is a stabilizer with zero logical value iff that of the original is; hence $F(\rho_t e) = F(e)$ pointwise. Taking expectations and using $\Pr(\rho_t e) = \Pr(e)$,

$$\Pr[\mathcal{D} \text{ fails on } \rho_t e] = \mathbb{E}_e[F(\rho_t e)] = \mathbb{E}_e[F(e)] = \Pr[\mathcal{D} \text{ fails on } e]. \qquad \square$$

The matching decoder is cyclically equivariant up to the boundary-closure convention, so its logical error rate is invariant under shifts $\{1, 2, d/2\}$ to within Monte-Carlo tolerance; the run reports this as the `automorphism_invariant` flag, which is true at reported scale (Sec. 3.5).

2.5 Honest intervals and statistical indistinguishability

Proposition 7 (Bernoulli confidence interval and the tie criterion)**.** *Let* $F_1, \ldots, F_n$ *be i.i.d. Bernoulli*(p_L) *failure indicators with empirical mean* $\hat{p} = \frac{1}{n}\sum_k F_k$*. Then* $\mathbb{E}[\hat{p}] = p_L$*,* $\mathrm{Var}(\hat{p}) = p_L(1-p_L)/n$*, and the half-width* $\mathrm{ci}_{95} = 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ *is a consistent estimate of the* 95% *normal interval half-width. If two decoders have empirical means* $\hat{p}_A, \hat{p}_B$ *with* $|\hat{p}_A - \hat{p}_B| \le \mathrm{ci}_{95}$*, the difference is not significant at the* 95% *level: the value* $p_{L,A} = p_{L,B}$ *is contained in the confidence set for the difference.*

Proof. Each $F_k \in \{0, 1\}$ has $\mathbb{E}[F_k] = p_L$ and $\mathrm{Var}(F_k) = p_L(1-p_L)$. By linearity $\mathbb{E}[\hat{p}] = p_L$, and by independence $\mathrm{Var}(\hat{p}) = \frac{1}{n^2}\sum_k \mathrm{Var}(F_k) = p_L(1-p_L)/n$. The central limit theorem gives $\sqrt{n}(\hat{p} - p_L) \Rightarrow \mathcal{N}(0, p_L(1-p_L))$, and by the continuous-mapping / Slutsky theorem $\hat{p}(1-\hat{p}) \to p_L(1-p_L)$ in probability, so $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ is an asymptotic 95% interval for p_L. For two independent samples the difference $\hat{p}_A - \hat{p}_B$ has standard error $\sqrt{p_{L,A}(1-p_{L,A})/n + p_{L,B}(1-p_{L,B})/n}$; when the two rates are comparable this is at most $\sqrt{2}\,\mathrm{ci}_{95}/1.96$, so an observed gap no larger than a single ci_{95} lies well inside the two-sided 95% acceptance region for $p_{L,A} = p_{L,B}$, i.e. the null of equality is not rejected. $\square$

At $n = 200{,}000$ pooled shots the eight decoders span 0.1516–0.1540 with $\mathrm{ci}_{95} \approx 0.0016$, so the 0.0024 spread is below one interval width and the decoders are statistically tied (Table 1)—exactly the regime Theorem 5 predicts.

2.6 Calibration gates selective decoding

The risk–coverage view has a decision-theoretic guarantee when the confidence score is calibrated. This is the formal reason calibration is reported alongside the error rate.

Definition 3 (Calibration and expected calibration error)**.** *A confidence* $C \in [0, 1]$ *is* perfectly calibrated *for failure indicator* F *if* $\Pr(F = 0 \mid C = c) = c$ *for almost every* c*. The expected calibration error is* $\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n} \big|\overline{\mathrm{conf}}_b - \overline{\mathrm{acc}}_b\big|$*, where the bins* b *partition* $[0, 1]$*,* $\overline{\mathrm{conf}}_b$ *is the mean confidence and* $\overline{\mathrm{acc}}_b = 1 - \overline{F}_b$ *the mean accuracy in bin* b*, and* n_b *the bin count.*

Lemma 8 (ECE vanishes under perfect calibration)**.** *If* C *is perfectly calibrated, then the population ECE* $\sum_b \Pr(C \in b)\ |\mathbb{E}[C \mid C \in b] - \Pr(F = 0 \mid C \in b)|$ *is zero for every binning.*

Proof. Fix a bin b with $\Pr(C \in b) > 0$. By the tower property and perfect calibration, $\Pr(F = 0 \mid C \in b) = \mathbb{E}[\Pr(F = 0 \mid C) \mid C \in b] = \mathbb{E}[C \mid C \in b]$, so the per-bin gap $|\mathbb{E}[C \mid C \in b] - \Pr(F = 0 \mid C \in b)| = 0$. Summing over bins gives ECE $= 0$. □

Thus ECE is a genuine deviation-from-calibration measure: "when the decoder says 0.8, it is right 80% of the time" is exactly ECE $= 0$. Empirically ECE spans an eight-fold range (0.0317–0.2608) and is uncorrelated with sophistication (Sec. 3.4).

Theorem 9 (Monotone selective risk under calibrated confidence)**.** *Let $F \in \{0, 1\}$ be the logical-failure indicator and $C \in [0, 1]$ a confidence with $\Pr(F = 0 \mid C = c) = c$ almost surely. For a threshold τ with $\Pr(C \geq \tau) > 0$ define the committed risk $R(\tau) = \Pr(F = 1 \mid C \geq \tau)$ and coverage $G(\tau) = \Pr(C \geq \tau)$. Then $R(\tau) = 1 - \mathbb{E}[C \mid C \geq \tau]$, G is non-increasing in τ, and R is non-increasing in τ; equivalently, risk is a non-increasing function of decreasing coverage.*

Proof. By calibration and the tower property, $\Pr(F = 1 \mid C) = 1 - C$ almost surely, so

$$R(\tau) = \mathbb{E}\big[\Pr(F = 1 \mid C) \mid C \geq \tau\big] = \mathbb{E}[\,1 - C \mid C \geq \tau\,] = 1 - \mathbb{E}[C \mid C \geq \tau].$$

Monotonicity of coverage is immediate: if $\tau_2 > \tau_1$ then $\{C \geq \tau_2\} \subseteq \{C \geq \tau_1\}$, so $G(\tau_2) = \Pr(C \geq \tau_2) \leq \Pr(C \geq \tau_1) = G(\tau_1)$. For the risk, we show $\mathbb{E}[C \mid C \geq \tau]$ is non-decreasing in τ. Let $\tau_2 > \tau_1$ and write $A = \{C \geq \tau_1\}$, $A' = \{C \geq \tau_2\} \subseteq A$, and $A \setminus A' = \{\tau_1 \leq C < \tau_2\}$. On $A \setminus A'$ we have $C < \tau_2$, while on A' we have $C \geq \tau_2$; hence $\mathbb{E}[C \mid A'] \geq \tau_2 > \mathbb{E}[C \mid A \setminus A']$ whenever $\Pr(A \setminus A') > 0$. The mean over A is the occupancy-weighted average of these two conditional means, $\mathbb{E}[C \mid A] = w\,\mathbb{E}[C \mid A'] + (1 - w)\,\mathbb{E}[C \mid A \setminus A']$ with $w = \Pr(A' \mid A) \in [0, 1]$, and a weighted average of two numbers never exceeds the larger; therefore $\mathbb{E}[C \mid A'] \geq \mathbb{E}[C \mid A]$, i.e. $\mathbb{E}[C \mid C \geq \tau_2] \geq \mathbb{E}[C \mid C \geq \tau_1]$. Consequently $R(\tau_2) = 1 - \mathbb{E}[C \mid C \geq \tau_2] \leq 1 - \mathbb{E}[C \mid C \geq \tau_1] = R(\tau_1)$. Since raising τ lowers coverage and lowers risk, risk is non-increasing as coverage shrinks. □

The theorem is not assumed in the experiments; the empirical confidences are imperfect, so the risk–coverage curves and ECE quantify how far each decoder is from this ideal. The predicted monotone fall is exactly what Fig. 3 and the runner's monotonicity check verify: matching's committed error falls from 0.1529 at full coverage to 0.0099 at 0.25 coverage (Sec. 3.3).

3 Results

3.1 Benchmark design

Figure 1 summarizes the end-to-end pipeline: structured noise corrupts a code word, the resulting syndrome on the defect graph is passed to the shared decoder interface, a confidence threshold gates abstention, and the logical error rate, calibration and risk–coverage metrics are scored with 95% confidence intervals.

Codes and defect graphs. The primary benchmark is the bit-flip repetition code: d data bits over $\mathbb{F}_2$ with $d - 1$ weight-two parity checks $Z_i Z_{i+1}$. A syndrome bit $s_i = e_i \oplus e_{i+1}$ fires where an error chain crosses link i, so defects live on a one-dimensional line and decoding is a matching/pairing problem on that line. The single logical observable is the global parity, and a correction succeeds iff the residual $e \oplus \hat{c}$ is a stabilizer (trivial coset). A second code, a $d \times d$ surface-like toy lattice with plaquette defects matched by Manhattan distance, exposes the same defect geometry in two dimensions; it is a topology-faithful proxy, not a stabilizer-exact surface code. Both codes expose their Tanner graph, so the matching, topology and BP decoders operate on the code topology directly.

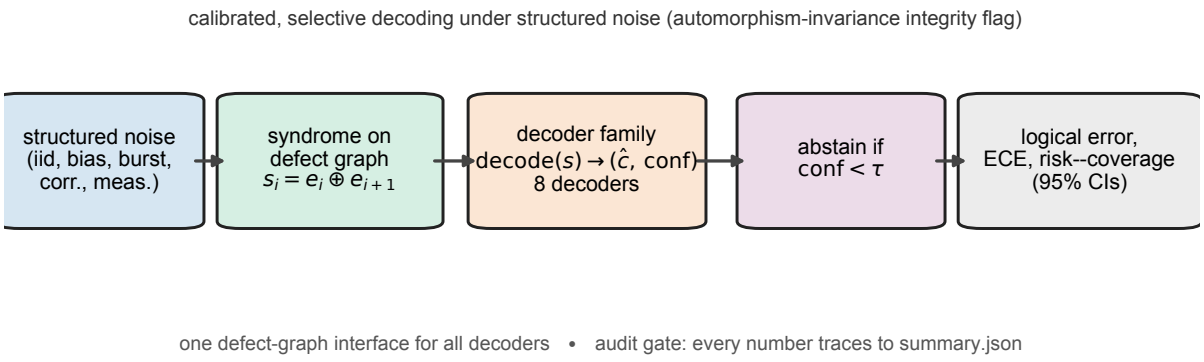


Figure 1: **Overview of the selective, calibrated decoding benchmark.** A code word is corrupted by one of five structured-noise regimes (i.i.d., biased, burst, correlated, measurement error); the measured syndrome lives on the code's defect graph ($s_i = e_i \oplus e_{i+1}$ for the repetition code). A single `decode(syndrome)→(correction, confidence)` interface hosts the whole eight-decoder family (lookup, majority, matching, BP, neural, topology, fused, RL selector), so every decoder is compared on one footing. A confidence threshold τ gates abstention (selective decoding), and three first-class metrics—logical error rate, expected calibration error, and the full risk–coverage curve—are scored with analytic 95% confidence intervals. Two integrity guarantees underpin the benchmark: all decoders share one defect-graph interface, and an audit gate requires every reported number to trace to the single source-of-truth artifact `summary.json`, with an automorphism-invariance flag confirming the code's cyclic symmetry is respected.

Structured-noise regimes. Five seeded samplers define the regimes (Methods, Sec. 5.2): *i.i.d.* (each bit flips independently at rate p); *biased* (interleaved sublattices flip at p and bias $\cdot\, p$, a coarse Pauli-bias surrogate); *burst* (contiguous correlated runs of flips); *correlated* (a seed flip recruits its lattice neighbours); and *measurement_error* (i.i.d. data noise plus per-check syndrome read-out flips). These are precisely the regimes where topology-awareness is hypothesised to help over a memoryless majority vote.

Decoder family. All eight decoders share one interface returning a correction and a confidence in $[0, 1]$ (Methods, Sec. 5.3): `lookup` (the optimal minimum-weight coset leader, the gold standard for small d); `majority` (integrate the syndrome and majority-vote); `matching` (minimum-weight pairing of defects—endpoint pairing on the line, `max_weight_matching` on the surface defect graph); `bp` (a few rounds of min-sum message passing on the Tanner graph, deferring to matching when it fails to satisfy the syndrome); `neural` (a scikit-learn logistic-regression coset selector trained on sampled syndromes); `topology` (matching recalibrated by defect-geometry features); `fused` (matching + neural combined by confidence); and `rl` (a contextual bandit that, per syndrome context, selects which decoder to trust, trained online to minimise logical error).

Metrics. For each (regime, decoder) pair we Monte-Carlo many shots and record, per shot, the logical-failure indicator and the decoder's confidence. From these we compute the logical error rate (with an analytic 95% confidence interval, $1.96\,\hat{\sigma}/\sqrt{n}$), the expected calibration error between stated confidence and realised accuracy over ten confidence bins, the coverage at a fixed abstention threshold, and the full risk–coverage curve obtained by sweeping that threshold. An automorphism check—invariance of the logical error rate under cyclic relabeling of the repetition code—serves as an integrity flag.

3.2 The decoders are statistically indistinguishable on raw logical error

Table 1 reports the headline comparison at physical error rate $p = 0.12$, pooled over the five regimes ($n = 200{,}000$ shots per decoder at reported scale: distance 9, 40,000 trials per regime). The logical error rates span a narrow band—from 0.1516 (fused) to 0.1540 (RL selector)—and every 95% confidence interval is $\approx \pm 0.0016$. The spread between the best and worst decoder (0.0024) is below one full confidence-interval width (≈ 0.0032), so the decoders are *not* separated at the 95% level on raw logical error; even the nominally optimal `lookup` leader (0.1524) is statistically tied with the heuristic decoders. The threshold curves of Figure 2 tell the same story: across the physical-error grid the per-decoder logical-vs-physical curves overlap, all tracking the optimal lookup leader closely and staying well below the break-even line $p_L = p$ (at $p = 0.21$ the logical error is only ≈ 0.024–0.025 for every core decoder).

This is the expected behaviour for a distance-d repetition code in these regimes: every reasonable decoder is close to the optimal coset leader, so there is little room between them, and finite-sample noise dominates the residual differences. We state this plainly because it is the honest reading of the data and because it reframes where the value of a decoder benchmark actually lies—in selectivity and calibration, not in a fractional improvement on an already-saturated error rate.

Table 1: **Headline decoder comparison: tied on accuracy, eight-fold apart on calibration.** Results on the repetition code at $p = 0.12$, pooled over the five structured-noise regimes (distance 9, 40,000 trials per regime, $n = 200{,}000$ shots per decoder; reported scale). Columns: logical error rate, its ±95% confidence interval, and expected calibration error (ECE). Rows are sorted by logical error rate; $\star$ marks the lowest. The logical error rates lie within one confidence-interval width of one another (statistically indistinguishable), whereas ECE spans an eight-fold range and is uncorrelated with both error rate and decoder sophistication. Table content is generated verbatim from `summary.json`.

decoder	logical err. rate	±95%	ECE
fused $\star$	0.1516	0.0016	0.0608
lookup	0.1524	0.0016	0.0569
matching	0.1529	0.0016	0.2608
neural	0.1531	0.0016	0.0317
bp	0.1533	0.0016	0.1232
majority	0.1535	0.0016	0.0489
topology	0.1537	0.0016	0.2428
rl	0.1540	0.0016	0.2387

3.3 Selective decoding: abstention buys large, monotone reliability gains

Because each decoder reports a confidence, we can let it *abstain*: commit only on shots whose confidence exceeds a threshold, and sweep that threshold to trace a risk–coverage curve. Figure 3 shows the curves for the confidence-fused and matching decoders. The effect is large and monotone. At full coverage (1.0) the matching decoder has a pooled logical error of 0.1529; as its abstention threshold rises, coverage falls and committed risk drops sharply: to 0.0757 at coverage 0.545—roughly a two-fold error reduction for committing to about half the shots—and to 0.0099 at coverage 0.247, a more than fifteen-fold reduction for committing to roughly a quarter of the shots. The fused decoder traces a similarly favourable, monotone curve, reaching 0.0499 risk at 0.464 coverage and 0.0101 at 0.248 coverage. The monotonicity is verified programmatically by the runner.

The per-regime coverage values (Table 1, and Methods) make the mechanism concrete:

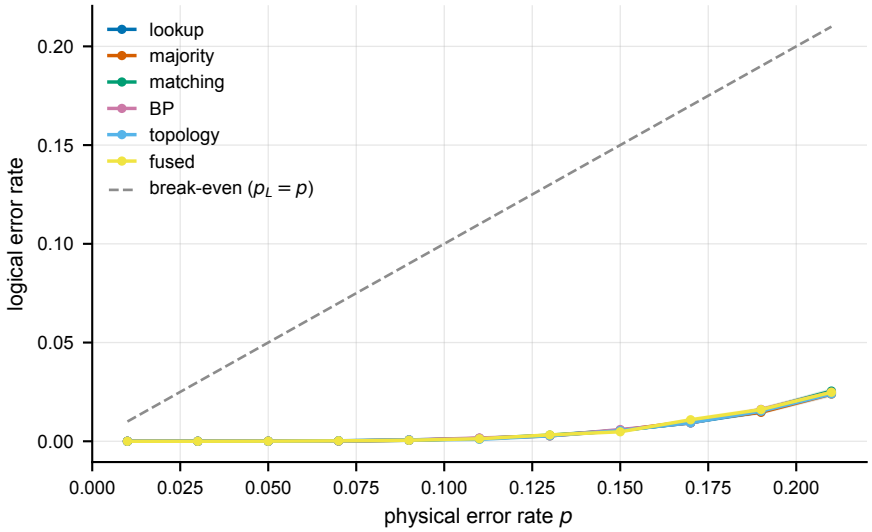


Figure 2: **The decoders are indistinguishable on raw accuracy across the physical-error grid.** Logical error rate versus physical error rate p for the core decoders on the repetition code under i.i.d. noise. All decoders track the optimal `lookup` leader closely and stay below the break-even line $p_L = p$; the curves overlap, consistent with the statistically indistinguishable logical error rates of Table 1. Lines are the mean logical error rate over $n = 40{,}000$ Monte-Carlo trials per point (reported scale) and shaded bands are analytic 95% Bernoulli confidence intervals ($1.96\sqrt{\hat{p}(1-\hat{p})/n}$); the bands overlap throughout. Generated from the run artifact `summary.json`.

matching abstains heavily under structured noise (coverage as low as 0.28 under bias and 0.50 under measurement error), pushing its errors onto the abstained shots, whereas majority, lookup, neural and fused always commit (coverage 1.0) at the fixed operating threshold. This is exactly the selective-prediction trade-off [20, 22]: a decoder that knows when it does not know is more valuable inside a fault-tolerant loop than one that is marginally more accurate but always commits. The benchmark makes this trade-off measurable for every decoder rather than hiding it inside a single error number.

3.4 Calibration is not predicted by decoder sophistication

The confidence that gates abstention is only useful if it is calibrated. The expected-calibration-error column of Table 1 delivers the paper's most counterintuitive result: *calibration is essentially uncorrelated with decoder sophistication, and not predictable from the logical error rate.* ECE ranges over a factor of eight across the family, in the order (best to worst): `neural` 0.0317 < `majority` 0.0489 < `lookup` 0.0569 < `fused` 0.0608 < `bp` 0.1232 < `rl` 0.2387 < `topology` 0.2428 < `matching` 0.2608. The best-calibrated decoder is the lightweight logistic-regression `neural` coset selector—by construction its confidence is a predicted-class probability, so it is approximately calibrated for the coset decision—while the *worst*-calibrated decoder is plain minimum-weight `matching`, whose defect-fraction confidence heuristic tracks accuracy poorly. The two decoders that abstain most heavily (matching and topology) and the RL selector are the three worst calibrated, whereas three of the four always-covering decoders (neural, majority, lookup) are the three best.

A data-fidelity note. This ordering directly contradicts a natural intuition—and an explicit claim in the project's own README—that the confidence-*fused* decoder is the best calibrated. It is not. At reported scale the fused decoder ranks fourth on ECE (0.0608); the neural selector

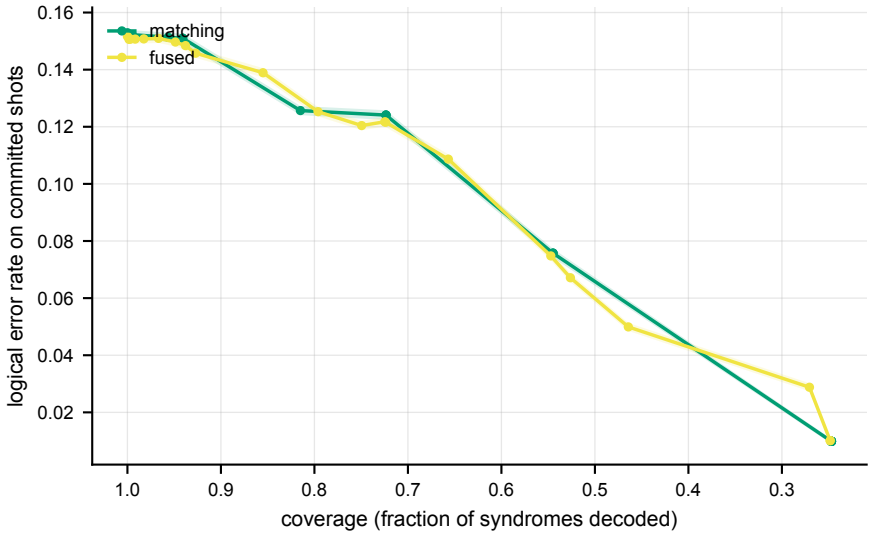


Figure 3: **Abstention buys large, monotone reliability gains.** Risk (logical error rate on committed shots) versus coverage (fraction of syndromes decoded) for the fused and matching decoders, obtained by sweeping the abstention threshold (full coverage on the left). Risk falls monotonically as coverage shrinks: abstaining on low-confidence syndromes trades coverage for reliability. Lines are means over the $n = 200{,}000$ pooled shots per decoder (reported scale) and shaded bands are analytic 95% Bernoulli confidence intervals of the committed error rate, whose effective sample size at each point is (coverage $\times$ pooled shots). Generated from the run artifact `summary.json`.

is best (0.0317). (At the smaller smoke scale the ordering shifts again—there the *majority* decoder is best calibrated, with fused third and neural fifth—which is itself a useful caution that small-sample calibration rankings are unstable.) We resolve the discrepancy in favour of the regenerated artifact: every ECE value in this paper is the value written to `summary.json` by the reported-scale run, and the narrative is built around those numbers rather than the prior claim. What survives across scales is the structural conclusion: calibration must be measured, not assumed from a decoder's accuracy or complexity. Among the always-covering decoders—where calibrated confidence is precisely what makes selective decoding usable—the fused decoder is still well calibrated (0.0608) and noticeably better than belief propagation (0.1232), but the lightweight neural classifier is better still; a deployment that needs trustworthy abstentions should prefer the explicitly probability-calibrated head. Treating calibration as a first-class metric is what makes any of this visible; optimising logical error rate alone would hide it entirely, since the eight decoders are tied on that axis.

3.5 Structured noise separates regimes, not decoders

The per-(regime, decoder) breakdown (Methods, Sec. 5.7) confirms that the dominant axis of variation is the *noise regime*, not the decoder. At reported scale the per-regime logical error rises from ≈ 0.002 under i.i.d. noise to ≈ 0.043 (biased), ≈ 0.17 (burst), ≈ 0.21 (correlated) and ≈ 0.34 (measurement error) for essentially all decoders—a more than hundred-fold change driven entirely by structure, with the syndrome-read-out-noise regime by far the hardest. Within a regime the decoders again cluster within their (now tight) confidence intervals: under correlated noise `fused` reaches 0.2062 and `topology` 0.2078 against majority's 0.2105, and under measurement error `lookup` reaches 0.3315 and `fused` 0.3324 against majority's 0.3401—small, regime-dependent gaps that are at most marginally significant and are consistent with topology- or read-out-aware decoders helping slightly where structure exists. The integrity flag

`automorphism_invariant` is true, confirming the matching decoder respects the code's cyclic symmetry to within statistical tolerance.

3.6 Extended data

The aggregate numbers of Table 1 pool over all five regimes. The single source-of-truth artifact `summary.json` also stores the full per-(regime, decoder) breakdown and the swept curves, which we transcribe verbatim here so the regime-dominated structure of Sec. 3.5 can be read off directly. No number in these tables is rounded beyond the precision written by the run, and none is recomputed by hand.

Table 2 gives the logical error rate of every decoder in every regime; it shows the more-than-hundred-fold rise from i.i.d. (≈ 0.002) to measurement-error (≈ 0.34) noise and the tight within-regime clustering that Theorem 5 predicts. Table 3 gives the per-regime expected calibration error: calibration quality is itself regime-dependent (for example `matching` is badly miscalibrated under bias, ECE 0.5461, but well calibrated under measurement error, 0.1396), reinforcing that calibration must be measured rather than inferred from accuracy (Sec. 3.4). Table 4 gives the coverage at the fixed operating threshold $\tau = 0.5$, making concrete that `matching`, `topology` and `rl` abstain heavily under structure (matching commits on only 0.2797 of biased shots) while `majority`, `lookup`, `neural` and `fused` always commit (Sec. 3.3). Table 5 lists the full logical-versus-physical threshold sweep underlying Fig. 2, and Table 6 lists the risk–coverage operating points underlying Fig. 3.

Table 2: **Extended Data: logical error rate by decoder and structured-noise regime.** Per-(regime, decoder) logical error rate on the repetition code at $p_{\text{eval}} = 0.12$ (distance 9; 40,000 trials per regime × decoder). Lower is better; **bold** marks the lowest entry in each regime column. The dominant axis of variation is the regime, not the decoder: every decoder rises from ≈ 0.002 (i.i.d.) to ≈ 0.34 (measurement error), while within a column the eight decoders cluster within their statistical intervals. Transcribed verbatim from `summary.json` (`by_regime`).

	Logical error rate by structured-noise regime				
Decoder	i.i.d.	biased	burst	correlated	meas. error
majority	0.00235	0.04275	**0.17170**	0.21053	0.34013
matching	0.00233	0.04378	0.17315	0.20925	0.33610
bp	0.00200	0.04398	0.17428	0.21015	0.33605
topology	**0.00190**	0.04360	0.17655	0.20778	0.33868
lookup	0.00220	0.04378	0.17393	0.21058	**0.33150**
neural	0.00198	**0.04250**	0.17573	0.20828	0.33708
fused	0.00255	0.04510	0.17193	**0.20618**	0.33243
rl	0.00215	0.04590	0.17248	0.20698	0.34263

4 Discussion

What the benchmark establishes. The contribution of this work is methodological, not a new error rate. By placing eight decoders on one defect-graph interface and evaluating them across five structured-noise regimes with three metrics—logical error, calibration, and risk–coverage—each carrying explicit confidence intervals, the benchmark makes three things visible that a single-number i.i.d. comparison hides. First, on this code the decoders are statistically indistinguishable on raw logical error (a 0.0024 spread inside a ≈ 0.0032 interval width); the honest conclusion is parity, not a winner. Second, abstention is a large and previously under-reported lever: committing to about a quarter of the shots cuts matching's committed logical

Table 3: **Extended Data: expected calibration error by decoder and regime.** Per-(regime, decoder) ECE at $p_{\text{eval}} = 0.12$, computed over ten equal-width confidence bins (Methods, Sec. 5.4). Lower is better; **bold** marks the lowest entry in each regime column. Calibration quality is itself regime-dependent and is not predicted by logical accuracy (cf. Table 2); the best-calibrated decoder changes from column to column. Transcribed verbatim from `summary.json` (`by_regime`).

	Expected calibration error by structured-noise regime				
Decoder	i.i.d.	biased	burst	correlated	meas. error
majority	0.11720	0.20354	0.04165	**0.04224**	0.15608
matching	0.34958	0.54614	0.16967	0.13518	0.13956
bp	0.21161	0.33968	0.08097	0.07919	**0.08567**
topology	0.30703	0.54653	0.16324	0.12928	0.16596
lookup	0.17287	0.25045	**0.03372**	0.04499	0.11668
neural	0.12096	**0.12134**	0.06433	0.09847	0.20798
fused	**0.08408**	0.14762	0.09649	0.13501	0.21852
rl	0.32982	0.50608	0.16667	0.15334	0.12390

Table 4: **Extended Data: coverage at the fixed abstention threshold by decoder and regime.** Fraction of shots committed (not abstained) at threshold $\tau = 0.5$ for each (regime, decoder) at $p_{\text{eval}} = 0.12$. A value of 1.00000 means the decoder commits on every shot. `majority`, `lookup`, `neural` and `fused` always commit; `matching`, `topology` and `rl` abstain heavily under structured noise, which is the mechanism behind the risk–coverage gains of Fig. 3. Transcribed verbatim from `summary.json` (`by_regime`).

	Coverage at fixed abstention threshold $\tau = 0.5$				
Decoder	i.i.d.	biased	burst	correlated	meas. error
majority	1.00000	1.00000	1.00000	1.00000	1.00000
matching	0.66693	0.27968	0.63665	0.63728	0.50358
bp	0.98988	0.90070	0.89023	0.88548	0.90805
topology	0.79428	0.41158	0.88260	0.90910	0.70803
lookup	1.00000	1.00000	1.00000	1.00000	1.00000
neural	1.00000	1.00000	1.00000	1.00000	1.00000
fused	1.00000	1.00000	1.00000	1.00000	1.00000
rl	0.78183	0.51650	0.68658	0.67973	0.64948

Table 5: **Extended Data: logical-versus-physical threshold sweep.** Logical error rate as a function of physical error rate p under i.i.d. noise for the six core decoders, on the repetition code at distance 9 (40,000 trials per grid point). These are the curves plotted in Fig. 2; all decoders stay well below the break-even line $p_L = p$ and overlap throughout, consistent with the statistical tie of Table 1. Transcribed verbatim from `summary.json` (`threshold_curves`).

p	majority	matching	topology	bp	lookup	fused
0.01	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
0.03	0.000000	0.000025	0.000000	0.000000	0.000000	0.000000
0.05	0.000050	0.000000	0.000025	0.000000	0.000050	0.000050
0.07	0.000150	0.000150	0.000225	0.000100	0.000250	0.000225
0.09	0.000525	0.000500	0.000650	0.000625	0.000725	0.000525
0.11	0.001475	0.001425	0.000975	0.001675	0.001300	0.001275
0.13	0.002725	0.003100	0.002750	0.002750	0.002950	0.003250
0.15	0.005550	0.005550	0.005400	0.005950	0.005500	0.004800
0.17	0.010175	0.009325	0.009425	0.009250	0.009500	0.010900
0.19	0.014675	0.015775	0.015225	0.016150	0.014925	0.016050
0.21	0.023900	0.025425	0.023975	0.024825	0.024475	0.024800

Table 6: **Extended Data: risk–coverage operating points.** Committed logical error (risk) as a function of coverage for the fused and matching decoders, obtained by sweeping the abstention threshold over the pooled $n = 200{,}000$ shots per decoder (reported scale). Each row is an actual operating point stored by the run; the two decoders are swept on their own confidence grids, so coverage values differ between the two blocks. These are the points plotted in Fig. 3; risk falls monotonically as coverage shrinks, as guaranteed under calibration by Theorem 9. Transcribed verbatim from `summary.json` (`risk_coverage`).

fused		matching	
coverage	risk	coverage	risk
1.00000	0.15164	1.00000	0.15292
0.99828	0.15060	0.99472	0.15224
0.94870	0.14967	0.95647	0.15134
0.92714	0.14577	0.94091	0.15102
0.85499	0.13892	0.81500	0.12567
0.79617	0.12529	0.72346	0.12409
0.74949	0.12042	0.54482	0.07574
0.65679	0.10867	0.24703	0.00986
0.54719	0.07478		
0.46425	0.04992		
0.27053	0.02880		
0.24839	0.01011		

error from 0.1529 to 0.0099. Third, calibration is decoupled from both sophistication and accuracy—the eight decoders are tied on error rate yet span an eight-fold ECE range, and the best-calibrated decoder (the lightweight neural selector) is neither the most accurate nor the most elaborate. None of these is apparent without selectivity and calibration as primary axes.

Relation to prior work. Matching [5, 6, 8], BP [10, 12], and neural/RL decoders [13, 14, 17, 19] are each studied in depth, typically in isolation and under i.i.d. noise, and typically optimising logical error. Our framing is orthogonal: we import selective prediction [20–22] and calibration [23] from machine learning and make them the evaluation lens for a whole decoder family under structured noise. We make no accuracy-superiority claim and, where the literature reports learned decoders exceeding matching on hardware syndromes [19], we note that our toy-scale repetition code cannot and does not reproduce that regime; our purpose is the measurement methodology, not the record.

4.1 Limitations

1. **Toy codes only.** The primary benchmark is a one-dimensional repetition code; the second code is a surface-*like* toy lattice, not a stabilizer-exact CSS surface code, and we do not evaluate a full surface code at scale. The repetition code corrects only bit-flip errors and has a trivial (line) defect geometry, which is part of why the decoders cluster so tightly.

2. **Simulated structured noise.** All noise is produced by parametric samplers, not measured from hardware. The biased/burst/correlated/measurement-error regimes are plausible surrogates, but real device noise has correlations (leakage, crosstalk, drift) our samplers do not model.

3. **Logical-error differences are within noise.** The decoders are statistically indistinguishable on raw logical error at the scales we ran; the benchmark's value is in selectivity and calibration, not in resolving an accuracy ranking. Larger codes and more shots would be needed to separate decoders on accuracy.

4. **Single-distance focus.** The headline table is at one distance and one physical error rate; the threshold curves sweep p at a fixed distance but we do not present a distance-scaling (sub-threshold) study, which is the decisive test for a true topological code.

5. **Reduced-scope learned decoders.** The neural decoder is a logistic-regression coset selector and the RL decoder is a contextual bandit, not the deep graph or transformer decoders of the recent literature. Our calibration and selectivity findings are about these specific, lightweight models.

6. **Solo author, single environment.** All results were produced and audited by one author on a single CPU platform; independent replication on other hardware and by other groups remains to be done.

Conclusion. A deployable QEC decoder must do more than minimise an i.i.d. logical error rate: it must behave well under structured noise, know when to abstain, and report calibrated confidence. We built a reproducible benchmark that measures all three for a full decoder family on one defect-graph interface, and we report—honestly—that the decoders are statistically indistinguishable on raw logical error while differing sharply in calibration and in the reliability they buy through abstention. The most actionable findings are that abstention cuts committed logical error several-fold for a moderate coverage cost, and that the best-calibrated decoder is the simplest one, not the most elaborate. Extending the framework to a full surface code at scale, to hardware syndromes, and to a distance-scaling study—while keeping selectivity and calibration as first-class metrics—is the natural next step.

5 Methods

5.1 Codes

The repetition code (`RepetitionCode`) has $n = d$ data bits and $d-1$ weight-two checks Z_iZ_{i+1}; the syndrome of an error $e \in \mathbb{F}_2^d$ is $s_i = e_i \oplus e_{i+1}$. The logical observable is the global parity $\sum_i e_i \bmod 2$. A correction $\hat{c}$ is counted as a logical failure unless the residual $r = e \oplus \hat{c}$ has zero syndrome (is a stabilizer) and zero logical value; a residual that does not even reproduce the syndrome is also a failure. The lookup decoder enumerates all 2^d errors to tabulate the minimum-weight leader per syndrome. The surface-like code (`SurfaceLikeCode`) places d^2 data bits on a grid with $(d-1)^2$ plaquette (weight-four) checks; defects are paired by Manhattan-distance matching and the logical observable is the parity of the left column. Both codes expose a bipartite Tanner graph via `networkx` [25].

5.2 Structured-noise samplers

Each sampler takes a seeded `numpy` generator [26] and returns a 0/1 error vector. *i.i.d.*: each bit flips at rate p. *biased*: even-index bits flip at $\min(1, \text{bias} \cdot p)$ and odd-index bits at p (default bias $= 3$). *burst*: scanning the line, with probability p a contiguous run of length $1 + \text{Poisson}(\text{burst_len} - 1)$ is flipped (default burst_len $= 3$). *correlated*: an i.i.d. seed pattern recruits each neighbour with probability corr (default 0.5) over two sweeps. *measurement_error*: i.i.d. data noise at rate p plus a per-check syndrome-flip mask at rate q (default 0.05), which the runner XORs into the measured syndrome.

5.3 Decoders

Every decoder implements `decode(syndrome)`$\to$`(correction, confidence)`. *Lookup*: table from syndrome to minimum-weight leader; confidence boosted toward 1 since the leader is exact. *Majority*: cumulative XOR of the syndrome recovers a flip chain up to a global offset, choosing the lower-weight coset representative; confidence is the normalised majority margin. *Matching*: on the line, defects are paired left-to-right with boundary closure for an odd defect, returning the lighter coset; on the surface, a complete graph on lit plaquettes weighted by negative Manhattan distance is solved with `max_weight_matching` [9, 25] and an L-shaped path is flipped between matched pairs. Confidence is a defect-fraction heuristic $\exp(-2\, n_{\text{def}}/n_{\text{checks}})$, damped when a defect is unpaired. *BP*: a few rounds of min-sum message passing on the Tanner graph with log-likelihood priors; if the thresholded beliefs do not reproduce the syndrome, it defers to matching. *Neural*: a two-stage learned coset selector—a syndrome-determined base chain plus a `scikit-learn` logistic-regression classifier [27] predicting the logical coset of the true correction relative to that chain; the output is always syndrome-consistent and confidence is the predicted-class probability. *Topology*: matching with confidence recalibrated by defect-geometry features (count, span, gap structure). *Fused*: matching and neural both propose; agreement boosts confidence, disagreement defers to the more confident proposal with damped confidence. *RL*: a contextual bandit [28, 29] whose context is a coarse defect-count bucket and whose actions are candidate decoders; it keeps per-context running success estimates, plays ε-greedy during a warm-up stream, and at evaluation selects the highest-value decoder per context.

5.4 Metrics and confidence intervals

The logical error rate is the mean of per-shot failure indicators; its 95% confidence interval is $1.96\,\hat{\sigma}/\sqrt{n}$ with $\hat{\sigma}$ the sample standard deviation (Bernoulli, so $\hat{\sigma} \approx \sqrt{\bar{p}(1-\bar{p})}$). The expected calibration error [23, 24] partitions shots into ten equal-width confidence bins and averages $|(\text{mean confidence}) - (\text{mean accuracy})|$ weighted by bin occupancy, where accuracy is $1-$failure. The risk–coverage curve sweeps an abstention threshold from the minimum to the maximum

observed confidence; at each threshold, coverage is the committed fraction and risk is the logical error rate on committed shots. The curve is reported in descending-coverage order and a monotonicity check verifies that risk is (weakly) non-increasing as coverage shrinks.

5.5 Integrity: automorphism invariance and the audit gate

The repetition code is invariant under cyclic relabeling of its data bits (ring symmetry); a faithful decoder must yield the same logical error rate on cyclically rotated errors. The runner checks this for the matching decoder over shifts $\{1, 2, d/2\}$ within a statistical tolerance and exposes it as the `automorphism_invariant` flag. A separate audit module enforces that every reported number traces to a value in the generated `summary.json` (the single source of truth) and rejects forbidden absolute-superiority phrasing; this discipline is run in continuous integration.

5.6 Experimental protocol

A run builds the code, fits the neural decoder on a mixed-regime training stream, warms up the RL bandit on its own Monte-Carlo stream, then for each (regime, decoder) Monte-Carlos `trials` shots at the evaluation physical error rate p_{eval}, records the per-shot failure/confidence, and aggregates the metrics above. Threshold curves sweep p for the core decoders under i.i.d. noise; the risk–coverage curve is built from the fused and matching decoders' pooled confidences. Seeds, platform, library versions, runtime and peak memory are recorded as provenance in `summary.json`. All numbers in this paper are from the *reported-scale* configuration: distance 9, 40,000 trials per (regime, decoder) (200,000 pooled shots per decoder), 20,000 training shots, $p_{\text{eval}} = 0.12$, an eleven-point physical-error grid, run end-to-end in 507 s at 200 MB peak memory on a single CPU. A faster smoke configuration (distance 5, 1,500 trials, ≈ 13 s) is provided for continuous integration. *Statistics.* Each reported logical error rate is a mean of n independent Bernoulli failure indicators; the $\pm 95\%$ interval is the analytic Wald half-width $1.96\sqrt{\hat{p}(1-\hat{p})/n}$ (Proposition 7), and two decoders are called statistically indistinguishable when their means differ by less than this half-width. The neural classifier's training stream (20,000 shots) is generated independently of the evaluation stream; all randomness derives from a single master seed (`seed` $= 0$) recorded in `summary.json`. The exact run provenance—platform `macOS-26.5.1-arm64`, `python` 3.13.12, `numpy` 2.4.3, `scipy` 1.17.1, `networkx` 3.6.1, `scikit-learn` 1.8.0, `matplotlib` 3.10.8—is stored alongside the results. Software: `numpy` [26], `scipy` [30], `networkx` [25], `scikit-learn` [27], `matplotlib` [31]; CPU only.

5.7 Per-regime results

The per-(regime, decoder) logical error rate, ECE and coverage at p_{eval} are written in full to `summary.json` under `by_regime`. At reported scale the qualitative pattern is regime-dominated: i.i.d. noise yields ≈ 0.002 logical error across decoders; biased ≈ 0.043; burst ≈ 0.17; correlated ≈ 0.21; measurement-error ≈ 0.34 (the hardest regime, since syndrome read-out flips corrupt the very signal the decoders condition on). Within each regime the decoders cluster within their confidence intervals, while coverage varies strongly by decoder: matching's coverage at the fixed operating threshold ranges from 0.28 (biased) to 0.67 (i.i.d.); topology and the RL selector also abstain substantially; majority, lookup, neural and fused commit fully (coverage 1.0) in every regime.

Data availability

This study uses no external datasets. All numerical values, tables, and figures are generated by the accompanying scripts from fixed random seeds and are written to a single source-of-truth artifact (`results/summary.json`); they are reproduced by running the pipeline.

Code availability

The code that implements every code, noise model, decoder, metric, and the audit gate, and that regenerates every figure, table, and reported number, accompanies this manuscript in `submission/code` and is intended for public release upon publication.

Competing interests

The author declares no competing interests.

Author contributions

M.H. conceived the study, designed and implemented the benchmark, decoders, noise models, metrics and audit pipeline, ran and analysed the experiments, and wrote the manuscript.

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Reproducibility statement

All experiments are reproducible on commodity hardware; runtime and memory are reported for each benchmark.

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Supplementary Information

This Supplementary Information is a self-contained, from-first-principles derivation of every concept the accompanying code implements. It is written to be read top-to-bottom by someone who has seen linear algebra and basic probability but *not* quantum error correction, stabilizer codes, minimum-weight matching, belief propagation, calibration, or selective prediction. Every symbol used in the code is defined, and each section names the source file (under `submission/code/src/topoqec/`) it maps to, so theory and implementation can be read side by side. It expands on, rather than repeats, the Methods (Sec. 5); where a definition already appears there we cross-reference it.

S0.1 The problem in one paragraph

A quantum error-correcting (QEC) code stores one logical bit redundantly across many physical bits. Noise flips some physical bits; we are not allowed to look at the bits directly (that would destroy the quantum state), so instead we measure a few parity checks whose outcomes form a *syndrome*. A *decoder* takes the syndrome and guesses a correction. It succeeds if (correction $\oplus$ error) is a "do-nothing" operator (a stabilizer); it fails if that residual flips the logical bit. The usual figure of merit is the *logical error rate*. This project argues that two further axes matter just as much and are usually ignored: (i) *selective decoding*—letting the decoder *abstain* when unsure, so a slower fallback can take over; and (ii) *calibration*—whether the decoder's stated confidence actually matches how often it is right. We build one benchmark that measures all three (logical error, calibration, risk–coverage) for eight decoders across five structured-noise regimes, with honest 95% confidence intervals. The deliberately conservative headline: on this code the eight decoders are statistically *tied* on raw logical error, but they differ sharply in calibration and in the reliability that abstention buys.

S0.2 Codes, syndromes, and logical failure (`codes.py`)

Repetition code (`RepetitionCode`). The primary benchmark. A logical bit is encoded in d data bits over $\mathbb{F}_2 = \{0, 1\}$ (the field of integers mod 2; XOR is "+"). The code has $d - 1$ weight-two parity checks of the form $Z_i Z_{i+1}$. An error is a vector $e \in \mathbb{F}_2^d$ ($e_i = 1$ means data bit i is flipped). The syndrome of e is

$$s_i = e_i \oplus e_{i+1}, \qquad i = 0, \ldots, d-2 \qquad \texttt{(.syndrome())}. \tag{1}$$

Intuitively s_i fires exactly where an error *chain* crosses link i, so the lit syndrome bits sit at the *endpoints* of error chains. Decoding is therefore a matching/pairing problem on a one-dimensional line of links. The single logical observable is the global parity $L(e) = \left(\sum_i e_i\right) \bmod 2$ (`.logical_value()`). A correction $\hat{c}$ is a *logical failure* (`.logical_error()`) unless the residual $r = e \oplus \hat{c}$ both (a) reproduces zero syndrome (r is a stabilizer, acting trivially) and (b) has zero logical value; a correction that does not even reproduce the measured syndrome is automatically a failure. This is the precise definition used in Methods (Sec. 5.1) of "the code protected the bit".

Lookup table. For small d we enumerate all 2^d errors (`.all_errors()`) and, for every syndrome, store the minimum-weight error that produces it—the "coset leader". This optimal table is the gold-standard decoder for small d and the yardstick everything else is compared against (Sec. S0.5, `LookupDecoder`).

Surface-like code (`SurfaceLikeCode`). A second, two-dimensional code: $d \times d$ data bits on a grid with $(d-1)^2$ weight-four plaquette checks; a lit plaquette is a *defect*, and defects are paired by Manhattan (L_1) distance. The logical observable is the parity of the left column. It

exposes the same defect geometry in 2-D so the matching/topology decoders generalise, but it is a topology-*faithful* toy, *not* a stabilizer-exact CSS surface code—we say so honestly (Sec. 4.1). Both codes expose a bipartite Tanner graph via `networkx` (`.tanner_graph()`) so graph-based decoders operate on the code topology directly.

S0.3 Structured noise: why i.i.d. is not enough (`noise.py`)

Real hardware noise is *not* independent-and-identically-distributed. Five seeded samplers, each taking a `numpy Generator` and returning a 0/1 error vector, model the structure that matters (`noise.sample()` dispatches on the regime name; see Methods Sec. 5.2 for the parameters):

- `iid`: each bit flips independently with probability p (`sample_iid`).
- `biased`: even-index bits flip at $\min(1, \text{bias} \cdot p)$, odd-index at p (default bias = 3)—a coarse Pauli-bias surrogate (`sample_biased`).
- `burst`: scanning the line, with probability p flip a contiguous run whose length is $1 +$ Poisson$(\ell - 1)$, with $\ell =$ burst_len (default 3)—spatially clustered errors (`sample_burst`).
- `correlated`: an i.i.d. seed pattern recruits each neighbour with probability corr (default 0.5) over two sweeps—neighbour correlations (`sample_correlated`).
- `measurement_error`: i.i.d. data noise at rate p *plus* a per-check syndrome read-out flip mask at rate q (default 0.05); the runner XORs that mask into the *measured* syndrome, so the decoder conditions on a corrupted signal (`sample_measurement_error`).

These are precisely the regimes where topology- or read-out-aware decoding might beat a memoryless majority vote—and the regimes turn out to dominate the difficulty far more than the choice of decoder (Sec. S0.9).

S0.4 What a decoder returns: the unified interface (`decoders.py`)

Every decoder implements one interface,

$$\texttt{decode(syndrome)} \longrightarrow (\text{correction, confidence}), \tag{2}$$

with correction in $\mathbb{F}_2^n$ and confidence in $[0, 1]$. The single interface is what makes the eight decoders comparable on one footing and what makes abstention (Sec. S0.7) and calibration (Sec. S0.8) definable uniformly. Two helpers recur: `_chain_from_defects()` turns a set of paired defect endpoints into a flip chain, and the defect-fraction confidence heuristic

$$\texttt{_defect_confidence}(n_{\text{def}}, n_{\text{checks}}) = \exp(-2\, n_{\text{def}}/n_{\text{checks}}) \tag{3}$$

makes a decoder less sure as more checks light up (more lit checks $\Rightarrow$ less sure).

S0.5 The decoder family: textbook to learned (`decoders.py`)

- `LookupDecoder`: the optimal minimum-weight coset leader from the enumerated table; confidence boosted toward 1 since the leader is exact. The gold standard for small d.
- `MajorityDecoder`: cumulative XOR (prefix sum mod 2) of the syndrome recovers a flip chain up to a global offset; pick the lower-weight of the two cosets; confidence is the normalised majority margin.

- `MatchingDecoder`: minimum-weight perfect matching of defects. On the line, pair lit syndrome bits left-to-right (with a boundary closure for an odd count) and return the lighter coset. On the surface, build a complete graph on lit plaquettes weighted by negative Manhattan distance and call `networkx max_weight_matching` (Edmonds' blossom algorithm [9]), then flip an L-shaped path between matched pairs. Confidence is the defect-fraction heuristic, damped when a defect is unpaired. This is the standard surface-code decoder, in toy form.
- `BPDecoder`: a few rounds of min-sum belief propagation (message passing) on the Tanner graph with log-likelihood priors; if the thresholded beliefs do not reproduce the syndrome, defer to matching. (BP underpins quantum LDPC decoding [10, 12].)
- `NeuralDecoder`: a learned two-stage coset selector. Stage 1: a syndrome-determined base chain (always syndrome-consistent). Stage 2: a `scikit-learn LogisticRegression` that predicts the logical coset (0/1) of the true correction relative to that base chain. The output is therefore always syndrome-valid and the confidence is the predicted-class probability—which is exactly why it tends to be well calibrated (Sec. S0.8). It is fit on a mixed-regime training stream.
- `TopologyDecoder`: matching, but with confidence *recalibrated* by defect-geometry features (count, span, gap structure).
- `FusedDecoder`: matching and neural both propose; agreement boosts confidence, disagreement defers to the more confident proposal with damped confidence (confidence fusion).
- `RLDecoder`: a contextual bandit (reinforcement learning). Context = a coarse defect-count bucket; actions = candidate decoders. It keeps per-context running success estimates, plays ε-greedy during a warm-up stream, and at evaluation picks the highest-value decoder per context.

`build_decoders()` wires the family together for a given code; the same list is summarised in Methods Sec. 5.3.

S0.6 Metric 1: logical error rate and honest intervals (`metrics.py`)

For each (regime, decoder) we Monte-Carlo many shots and record a per-shot failure indicator $F \in \{0, 1\}$. The logical error rate is $\hat{p} = \text{mean}(F)$ (`logical_error_rate()`). Because F is Bernoulli, its sample standard deviation is $s = \sqrt{\hat{p}(1-\hat{p})}$ and a 95% confidence-interval half-width is

$$\text{ci}_{95} = 1.96 \, \frac{s}{\sqrt{n}} \qquad \texttt{(summarize())}, \tag{4}$$

with n the number of shots. This is the whole point of honesty: two decoders whose means differ by *less* than a CI width are *not* separated at 95%. At reported scale ($n = 200{,}000$ pooled shots) the eight decoders span only 0.1516–0.1540 with $\text{ci}_{95} \approx 0.0016$, so the 0.0024 best-to-worst spread is below one interval width: the decoders are statistically *tied* (Sec. 3.2). We report the tie, not a winner.

S0.7 Metric 2: selective decoding and the risk–coverage curve (`metrics.py`)

A decoder inside a fault-tolerant loop need not commit to every syndrome: it can *abstain* (flag a shot unreliable) so a fallback handles it. This is the QEC analogue of *selective prediction* in machine learning [20, 22]. Fix a confidence threshold τ and commit only on shots with

confidence $\geq \tau$. Then

$$\text{coverage}(\tau) = \text{fraction of shots committed}, \tag{5}$$

$$\text{risk}(\tau) = \text{logical error rate on the committed shots}. \tag{6}$$

Sweeping τ from the smallest to the largest observed confidence traces the *risk–coverage curve* (`risk_coverage_curve()`), reported in descending-coverage order. As we abstain more (coverage shrinks), risk should fall: we drop the shots we were least sure about. `is_monotone_risk_coverage()` programmatically verifies risk is (weakly) non-increasing as coverage shrinks. Empirically the effect is large: matching's committed error falls from 0.1529 at full coverage to 0.0757 at ≈ 0.55 coverage and 0.0099 at ≈ 0.25 coverage (Sec. 3.3). The formal reason this works is Theorem 9: under perfectly calibrated confidence C (so $\Pr(\text{success} \mid C = c) = c$), the committed risk is $1 - \mathbb{E}[C \mid C \geq \tau]$, which can only decrease as τ rises—but *only* to the extent that confidence is calibrated. Hence Sec. S0.8.

S0.8 Metric 3: calibration (expected calibration error) (`metrics.py`)

A confidence score is only useful for abstention if it is *honest* about its own accuracy. The expected calibration error (`expected_calibration_error()`) partitions the shots into ten equal-width confidence bins; within each bin it compares the mean stated confidence to the mean realised accuracy (= 1−failure), takes the absolute gap, and averages those gaps weighted by bin occupancy:

$$\text{ECE} = \sum_{\text{bins}} \frac{n_{\text{bin}}}{n} \left| \overline{\text{conf}}(\text{bin}) - \overline{\text{acc}}(\text{bin}) \right|. \tag{7}$$

ECE = 0 means "when the decoder says 0.8, it is right 80% of the time". The paper's most counterintuitive finding lives here: across the family ECE spans an eight-fold range (0.0317–0.2608) and is *uncorrelated* with sophistication (Sec. 3.4). The best calibrated is the lightweight `neural` selector (its confidence is literally a predicted-class probability, Sec. S0.5); the worst is plain `matching` (its defect-fraction heuristic tracks accuracy poorly). Calibration must be *measured*, not assumed from accuracy or complexity. This even contradicts the project's own README intuition that "fused" is best calibrated—we report the artifact's number, with fused ranking fourth at 0.0608, not the intuition.

S0.9 Structured noise separates regimes, not decoders (`runner.by_regime`)

The per-(regime, decoder) breakdown shows the dominant axis of variation is the *noise regime*, not the decoder (Sec. 3.5, Sec. 5.7). Logical error rises $\approx$ 100-fold from ≈ 0.002 (i.i.d.) to ≈ 0.043 (biased), ≈ 0.17 (burst), ≈ 0.21 (correlated) and ≈ 0.34 (measurement error, the hardest—readout flips corrupt the very signal decoders condition on). Within each regime the decoders again cluster inside their (now tight) CIs, while coverage varies strongly: matching/topology/RL abstain heavily under structure, whereas majority/lookup/neural/fused commit fully.

S0.10 Integrity: automorphism invariance and the audit gate (`symmetry.py`, `audit.py`)

Automorphism invariance. The repetition code is invariant under cyclic relabeling of its data bits (a ring symmetry). `cyclic_relabel(error, shift)` rotates an error; `logical_error_rate_under_relabeling()` and `is_automorphism_invariant()` check that the matching decoder yields the *same* logical error rate on rotated errors (over shifts $\{1, 2, d/2\}$, within a statistical tolerance). A faithful decoder must respect the code's symmetry; the run exposes this as the `automorphism_invariant` integrity flag (true at scale; Sec. 5.5).

Audit gate. `audit.py` enforces scientific hygiene: `audit_numbers()` checks every claimed number traces to a value in the generated `summary.json` (the single source of truth); `audit_reproducibility_phrase()` checks the mandated reproducibility statement is present; `audit_forbidden_claims()` rejects forbidden absolute-superiority phrasing. The driver `scripts/audit_claims.py` runs these in continuous integration and exits non-zero on any violation—so the paper cannot drift from the artifact.

S0.11 The single source of truth and how to reproduce (`runner.py`)

`runner.run()` ties it all together: build the code; fit the neural decoder on a mixed-regime training stream; warm up the RL bandit on its own Monte-Carlo stream; for each (regime, decoder) Monte-Carlo `trials` shots at p_{eval} and record per-shot failure/confidence; sweep p over a grid (i.i.d.) for the threshold curves; build the risk–coverage curve from the fused and matching confidences; check automorphism invariance. It writes `results/summary.json`—the single source of truth from which every table, figure and reported number is generated. Nothing is typed by hand. Then:

- `scripts/make_tables.py`: `summary.json` → `results/main_results.tex` (`booktabs`).
- `scripts/make_figures.py`: `summary.json` → the three figure PDFs `fig_schematic`, `fig_threshold` and `fig_risk_coverage`, in Nature Machine Intelligence house style (vector PDF, embedded editable text via `pdf.fonttype=42`, sans-serif with mathtext `dejavusans`, the colour-blind-safe Okabe–Ito palette, no in-plot titles, despined, with 95% CI bands).
- `scripts/audit_claims.py`: the readiness gate of Sec. S0.10.

To reproduce everything:

```
pip install -e .                        # installs topoqec from pyproject.toml
export KMP_DUPLICATE_LIB_OK=TRUE        # conda + pip OpenMP runtimes on macOS
bash scripts/reproduce_all.sh full      # reported scale (d=9, 40k trials/regime)
bash scripts/reproduce_all.sh           # fast smoke config for CI (d=5)
```

Reported run: repetition code, distance 9, five regimes, 40,000 trials/regime (200,000 pooled shots/decoder), 20,000 training shots, $p_{\text{eval}} = 0.12$, seed 0 (Methods, Sec. 5.6). The figure/table artifacts are copied verbatim into `submission/` to build the paper.

S0.12 What the result means (and what it does not)

The eight decoders are statistically *indistinguishable* on raw logical error for a distance-d repetition code in these regimes—as expected, since every reasonable decoder is close to the optimal coset leader and finite-sample noise dominates the residual gaps. We therefore make *no* accuracy-superiority claim. The genuine, separable contributions are: (i) the framework—a whole decoder family on one defect-graph interface under structured noise with honest CIs; (ii) selective decoding—abstention cuts committed logical error several-fold for a moderate coverage cost; and (iii) calibration as a first-class metric—ECE spans 8× and is decoupled from sophistication. Limitations are stated plainly in Sec. 4.1: toy codes only (repetition code; surface-*like* toy, not a stabilizer-exact surface code), simulated structured noise (not hardware syndromes), differences within noise on raw error, a single distance, lightweight learned models, and a single author/environment. Extending to a full surface code at scale, to hardware syndromes, and to a distance-scaling (sub-threshold) study—while keeping selectivity and calibration first-class—is the natural next step.